

Mathematics 418 — Midterm — 45 minutes

3rd September 1752

- The test consists of 10 pages and 7 questions worth a total of 28 marks.
- This is a closed-book examination. **None of the following are allowed:** documents, cheat sheets or electronic devices of any kind (including calculators, phones, smart watches, etc.)
- No work on this page will be marked.
- Fill in the information below before turning to the questions.

싸이	
Signature	
Name	

demo version 1

Please do not write on this page — it will not be marked.

Additional instructions

- Please use the spaces indicated.
- If you require extra paper then put up your hand and ask your instructor.
 - You must put your name and student number on any extra pages.
 - You must indicate the test-number and question.
 - Please do this **on both sides** of any extra pages.
- Please do not dismember your test. You must submit all pages.
- Smoking is strictly prohibited during the test.

Formula sheet

You may find the following formulas useful in the questions that follow.

$$a^2 + b^2 = c^2$$

$$e^{i\pi} + 1 = 0$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\frac{d}{dx}(f+g) = \frac{df}{dx} + \frac{dg}{dx}$$

$$\int (f+g)dx = \int fdx + \int gdx$$

$$\cos(a+b) = \cos a \cos b - \sin a \sin b$$

$$\sin(a+b) = \sin a \cos b + \cos a \sin b$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

Q1. 6 marks Compute the following limits.

(a) $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

Answer:

(b) $\lim_{x \rightarrow +\infty} \frac{3x + 2}{\sqrt{x^2 + 4x - 7}}$

Answer:

(c) $\lim_{x \rightarrow 0^+} \frac{x}{|x|}$

Answer: